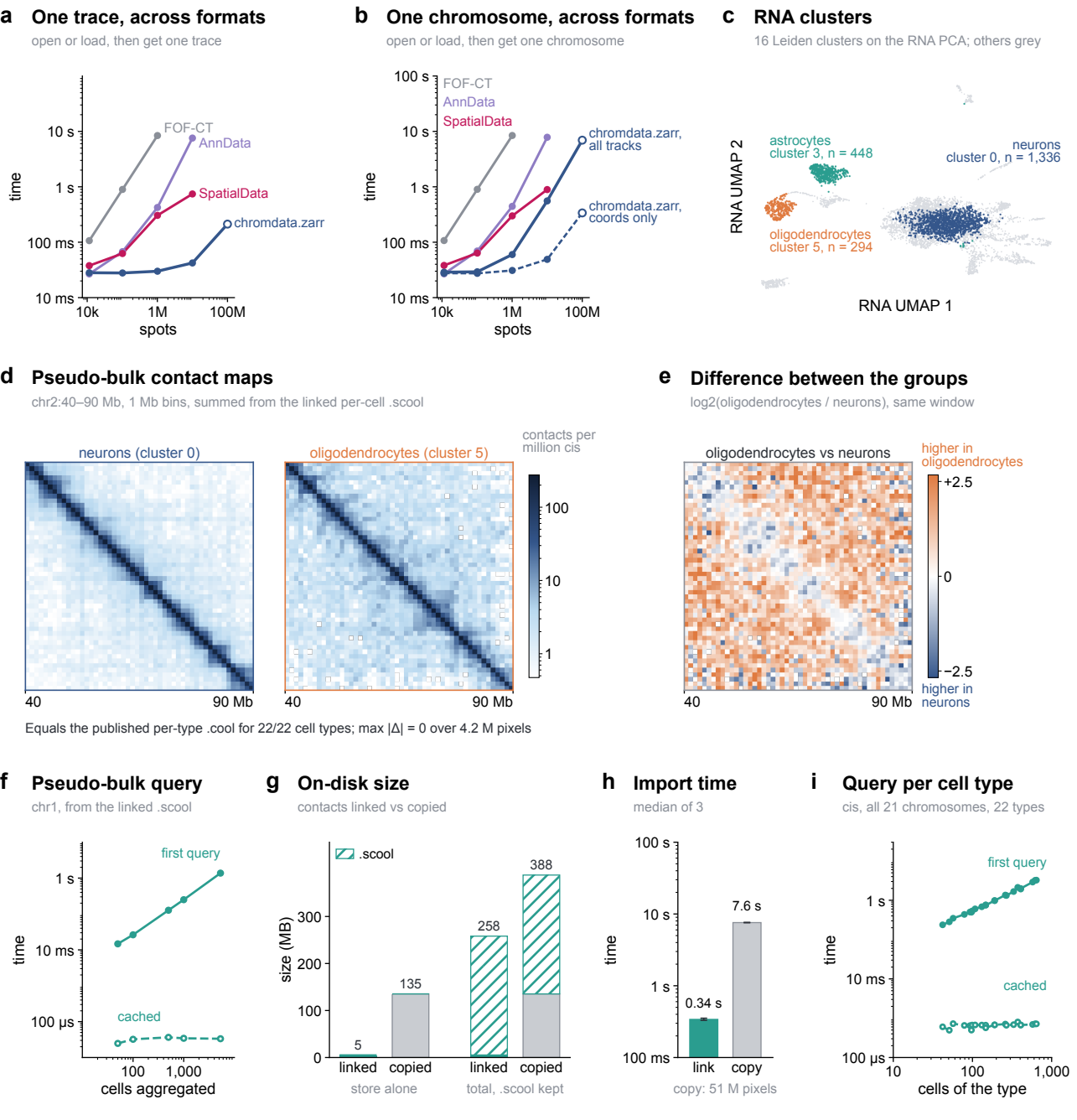

a
One trace, across formats
open or load, then get one trace
time
10 s
1 s
100 ms
10 ms
FOF-CT
AnnData
SpatialData
chromdata.zarr
10k
1M
100M
spots
b
One chromosome, across formats
open or load, then get one chromosome
time
100 s
10 s
1 s
100 ms
10 ms
FOF-CT
AnnData
SpatialData
chromdata.zarr, all tracks
chromdata.zarr, coords only
10k
1M
100M
spots
c
RNA clusters
16 Leiden clusters on the RNA PCA; others grey
RNA UMAP 2
RNA UMAP 1
astrocytes
cluster 3, n = 448
neurons
cluster 0, n = 1,336
oligodendrocytes
cluster 5, n = 294
d
Pseudo-bulk contact maps
chr2:40–90 Mb, 1 Mb bins, summed from the linked per-cell .scool
neurons (cluster 0)
oligodendrocytes (cluster 5)
contacts per million cis
100
10
1
40
90 Mb
40
90 Mb
Equals the published per-type .cool for 22/22 cell types; max |Δ| = 0 over 4.2 M pixels
e
Difference between the groups
log2(oligodendrocytes / neurons), same window
oligodendrocytes vs neurons
higher in oligodendrocytes
+2.5
0
−2.5
higher in neurons
40
90 Mb
f
Pseudo-bulk query
chr1, from the linked .scool
time
1 s
10 ms
100 µs
first query
cached
100
1,000
cells aggregated
g
On-disk size
contacts linked vs copied
.scool
size (MB)
300
200
100
0
5
135
258
388
linked
copied
linked
copied
store alone
total, .scool kept
h
Import time
median of 3
time
100 s
10 s
1 s
100 ms
0.34 s
7.6 s
link
copy
copy: 51 M pixels
i
Query per cell type
cis, all 21 chromosomes, 22 types
time
1 s
10 ms
100 µs
first query
cached
10
100
1,000
cells of the type